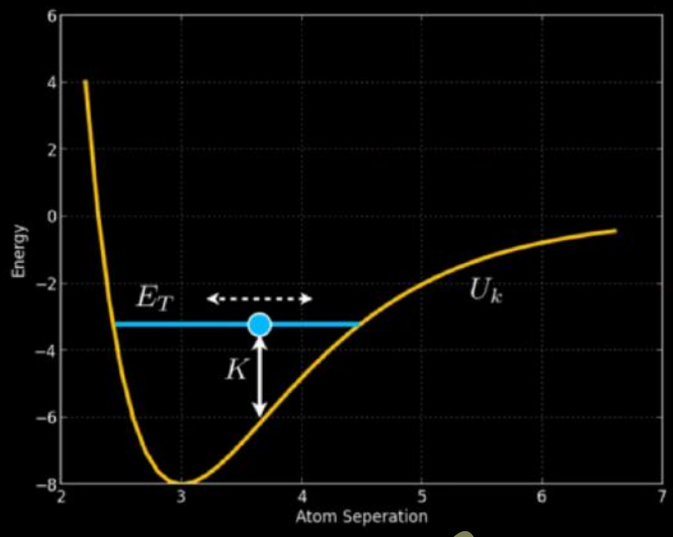
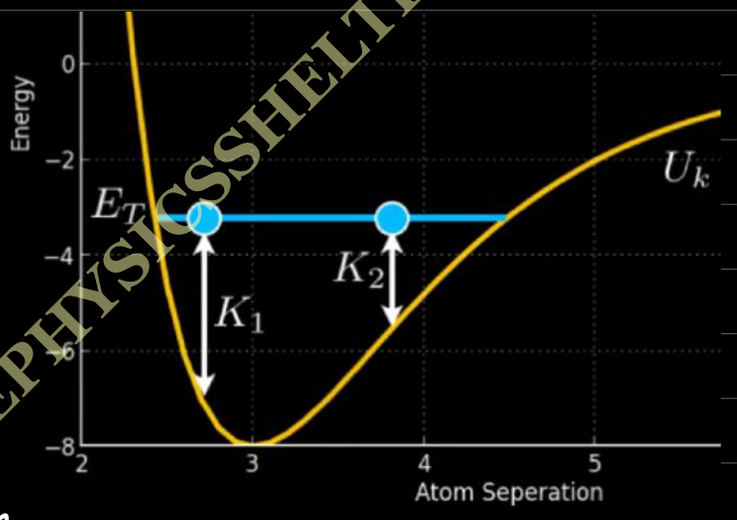
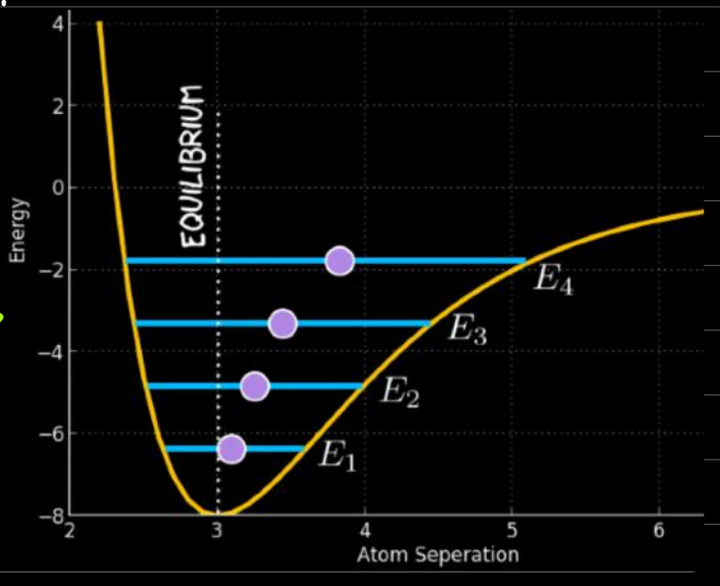
Cause Of Thermal Expansion:

The adjacent figure shows variation of potential energy vs interatomic spacing for a typical material. This graph is asymmetric about equilibrium position, At absolute zero, the energy of oscillation is negligible and particles stay at equilibrium, however as temperature increases, the interatomic spacing begins to oscillate in large amplitude. The particles spend more time at larger spacings (due to lower KE) than at smaller spacings (higher KE as clear from graph). Thus average spacing between atoms increases with increase in thermal energy. This causes thermal expansion at microscopic level and hence producing proportional macroscopic effects. Imagine if you have a model of a house, and you increase spacing between every two elements by same factor then the enlarged model will be still similar to original model in all respects, but just magnified by same scale factor in all directions.







→ Thus thermal expansion is just like a 3 Dimensional photographic (stereographic) enlargement.

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Coefficient Of Linear Thermal Expansion (α):

As discussed in previous section, material expands on heating. Let a rod of length l_0 expand by an amount Δl upon increasing the temperature from T to $T+\Delta T$, then we define $\Delta l = l_0 \alpha \Delta T$ or $\alpha = \frac{1}{\Delta T}\frac{\Delta l}{l_0}$. Thus we can define α or coefficient of linear thermal expansion as fractional change in length per unit rise in temperature. For most materials, it can be considered as constant for small temperature ranges however, it could also be a function of temperature, then we can define instantaneous coefficient of thermal expansion as :

$$\alpha = \frac{1}{l}\frac{dl}{dT} \quad \text{where } l \text{ is instantaneous length.}$$

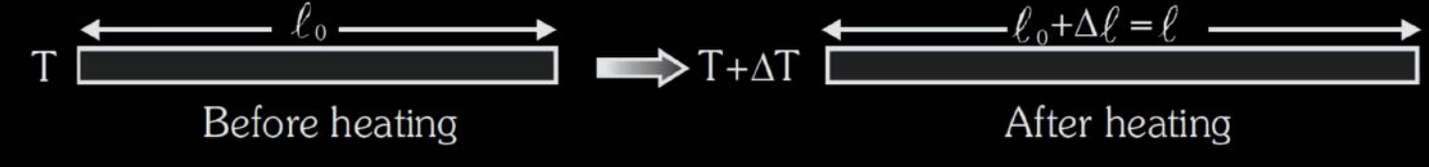
If α is constant:

$$\int_{l_0}^{l}\frac{dl}{l} = \int_{T_i}^{T_f}\alpha\, dT \Rightarrow l = l_0 e^{\alpha\Delta T}$$

Usually α is of the order of 10^{-4}, so $\alpha\Delta T << 1$ for for temperature increases of order 10^2. Also recall that

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} \ldots \approx 1 + x \text{ for } x << 1$$

$$\Rightarrow \boxed{l = l_0(1+\alpha\Delta T)} \leftarrow \text{Only if } \alpha\Delta T << 1 \text{ \& } \alpha = \text{Constant.}$$



→ If $\alpha\Delta T$ is not small or α is not constant then for finding final length we will need to integrate the following DE:

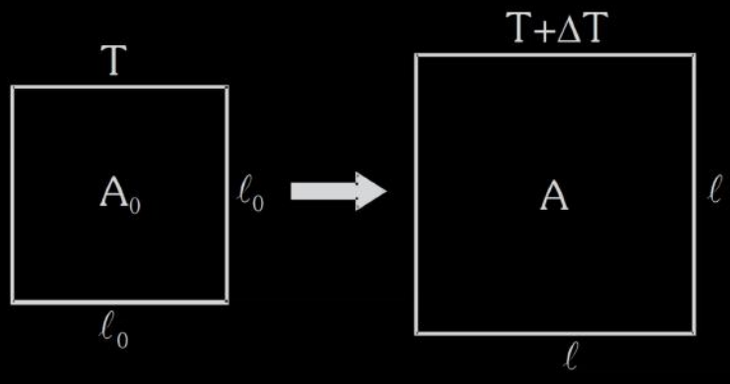
$$\alpha = \frac{1}{l}\frac{dl}{dT} \Rightarrow \ln\left(\frac{l}{l_0}\right) = \int_{T_i}^{T_f}\alpha\, dT$$

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Coefficient Of Superficial (Areal) Expansion (β):

$A = A_0(1+\beta\Delta T)$

$\Rightarrow \beta = \frac{1}{\Delta T}\left(\frac{\Delta A}{A_0}\right)$



Thus β is defined as fractional change in area per unit rise in temperature.

Relation between α & β: $A_0 = \ell_0^2$, $A = [\ell_0(1+\alpha\Delta T)]^2$

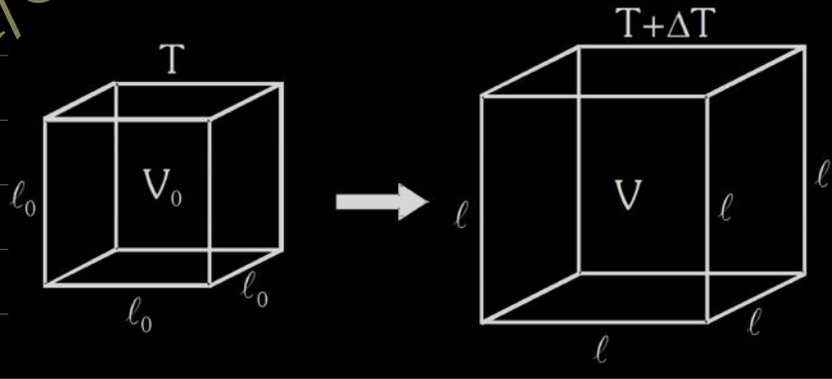
$\Rightarrow A = A_0[1+2\alpha\Delta T]$ if $\alpha\Delta T << 1$

$\Rightarrow \boxed{\beta = 2\alpha}$

Coefficient Of Cubical (Volumetric) Expansion (γ):

$V = V_0(1+\gamma\Delta T)$

$\Rightarrow \gamma = \frac{1}{\Delta T}\left(\frac{\Delta V}{V_0}\right)$



Thus coefficient of cubical expansion is fractional change in Volume per unit rise in Temperature.

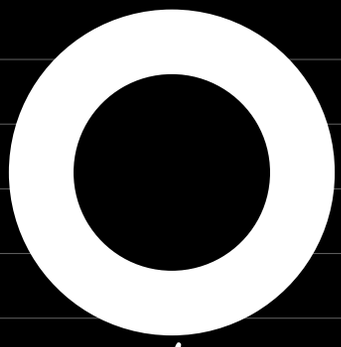
Relation Between α & γ: $V_0 = \ell_0^3$, $V = [\ell_0(1+\alpha\Delta T)]^3$

$\Rightarrow V = V_0[1+3\alpha\Delta T]$ if $\alpha\Delta T << 1$

$\Rightarrow \boxed{\gamma = 3\alpha}$

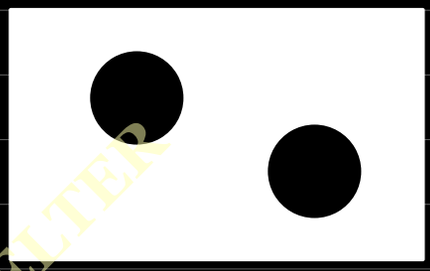
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EXAMPLE: A solid object has a hole in it. What happens to the size of the hole when the temperature of the object is increased.



Solution: Recall that Thermal expansion is a phenomenon at microscopic level. ∴ the diameter of hole will increase.

Example: A metal sheet has two holes as shown in the figure. What happens to separation between holes when the sheet is heated?



Ans: The separation would increase (photographic enlargement).

CHANGE IN TIME PERIOD OF A PENDULUM:

Recall that $T = 2\pi\sqrt{\frac{l}{g}} \Rightarrow \frac{\Delta T}{T} = \frac{1}{2}\frac{\Delta l}{l} = \frac{1}{2}\left[\frac{l\alpha\Delta\theta}{l}\right]$

Here θ is temperature and T is time. $\Rightarrow \frac{\Delta T}{T} = \frac{1}{2}\alpha\Delta\theta$

$\Rightarrow$ $\boxed{\Delta T = \frac{1}{2}T\alpha\Delta\theta}$

CALCULATING TRUE LENGTH FROM AN EXPANDED SCALE:

Q. A metre scale graduated at 0°C reports a length to be l_0 at a temperature $\Delta\theta$. What is the true length?

Sol: Let the true length be L and least count at 0°C be m, $\Rightarrow$ Required numerical value: $l_0 = \frac{L}{m}$ units

Reported $l_\theta = \frac{L}{m(1+\alpha\Delta\theta)}$ units $= \frac{l_0}{(1+\alpha\Delta\theta)}$

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$\Rightarrow$ $\boxed{l_0 = l_0(1+\alpha\Delta\theta)}$

CHANGE IN A CONTAINER'S CAPACITY ON HEATING:

Since thermal expansion is like 3D photographic enlargement $\therefore$ the container's capacity also gets scaled up by the same factor as the volume of its material. If V_0 is the container's capacity before heating and V is the capacity after heating then:

$$\boxed{V = V_0(1+\gamma_m \Delta T)}$$

Here γ_m is γ of the material of the container.

EFFECT OF TEMPERATURE ON DENSITY OF SUBSTANCE:

$$\rho_0 = \frac{m}{V_0}, \quad \rho = \frac{m}{V_0(1+\gamma\Delta T)} = \rho_0(1+\gamma\Delta T)^{-1}$$

$\Rightarrow$ $\boxed{\rho \simeq \rho_0(1-\gamma\Delta T)}$

EFFECT OF TEMPERATURE ON BUOYANT FORCE (Fully Submerged Body):

$$F_B = \rho_0 V_0 g, \quad F_B' = \rho_0(1-\gamma_L\Delta T)\, V_0(1+\gamma_s\Delta T)\, g$$

$\gamma_L \rightarrow$ For liquid, $\gamma_s \rightarrow$ For solid, $\rho_0 \rightarrow$ Liquid density.

Neglecting Second Order terms in F_B': $\frac{F_B'}{F_B} = 1+(\gamma_s-\gamma_L)\Delta T$

Usually, $\gamma_s < \gamma_l \Rightarrow F_B' < F_B$.

Reduction in apparent weight: $F_B' - F_B = \rho_0 V_0 g\,[(\gamma_s-\gamma_L)\Delta T]$

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Condition For Maintaining Constant Liquid Height:

For Liquid, $V_0 = A_0 h_0$

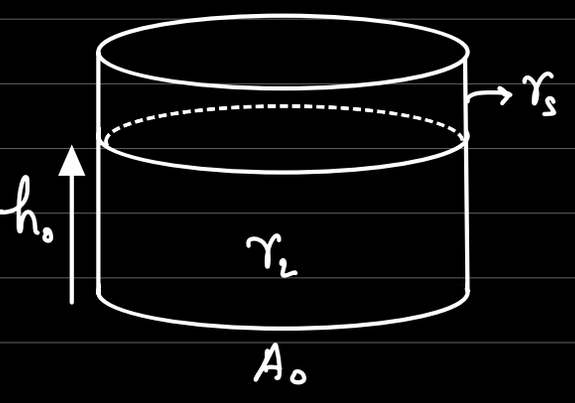
New Volume of liquid:

$$V = V_0(1+\gamma_L \Delta T)$$

New Vessel Base Area $A = A_0(1+\beta_s \Delta T)$

$\Rightarrow$ New Height $= V/A$

$$\Rightarrow h = \frac{V_0}{A_0}(1+\gamma_L \Delta T)(1+\beta_s \Delta T)^{-1}$$



$$\Rightarrow h = h_0(1+(\gamma_L-\beta_s)\Delta T)$$ Neglecting Second order terms.

$\Rightarrow$ If $\gamma_L > \beta_s$ Height Increases (or If $\gamma_L \geq \frac{2}{3}\gamma_s$, $h \geq h_0$)

If $\gamma_L < \frac{2}{3}\gamma_s \Rightarrow h$ decreases

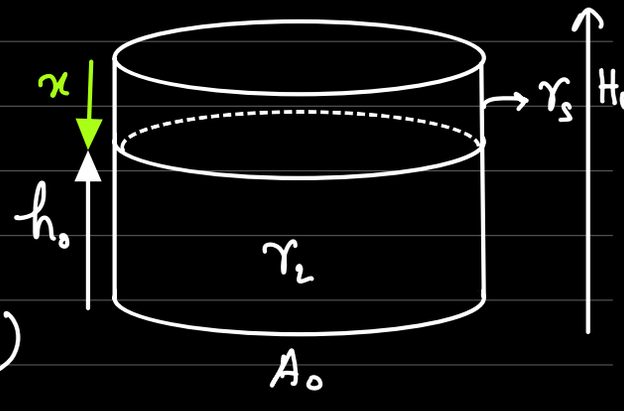
If $\boxed{\gamma_L = \frac{2}{3}\gamma_s} \Rightarrow h$ remains Same

Condition For Maintaining Constant Height Of Air Column:

Let V_0 = Volume of Vessel.

$$\Rightarrow x = \frac{V_0}{A_0} - h_0$$

Similarly $x' = \frac{V}{A} - h$



$$\Rightarrow x' = \frac{V_0(1+\gamma_s \Delta T)}{A_0(1+\beta_s \Delta T)} - h_0(1+(\gamma_L-\beta_s)\Delta T)$$

$$\Rightarrow x' = \frac{V_0}{A_0}[1+(\gamma_s-\beta_s)\Delta T] - h_0(1+(\gamma_L-\beta_s)\Delta T)$$

$$x' = x + \left[\frac{V_0}{A_0}\alpha_s \Delta T - h_0(\gamma_L-\beta_s)\Delta T\right]$$

$$\Rightarrow x' = x_0 + \Delta T[H_0 \alpha_s - h_0(\gamma_L - 2\alpha_s)]$$

If $x' = x_0 \Rightarrow H_0 \alpha_s = h_0(\gamma_L - 2\alpha_s)$ Ans.

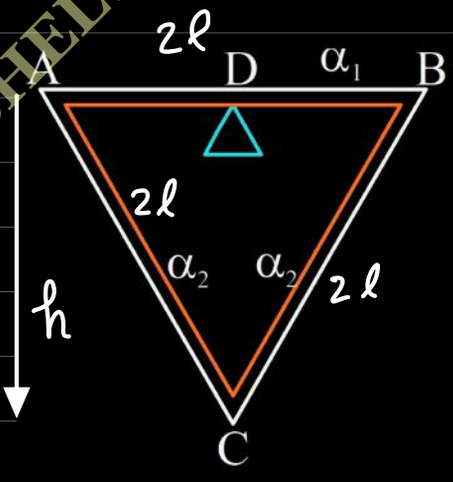
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EXAMPLE: If fraction of a solid remains submerged in a certain liquid at a temperature T_0, what fraction will be submerged when temperature increases by ΔT, given γ_s and γ_l are γ for solid and liquid phases.

Sol: $f = \frac{\rho_s}{\rho_l}$, $f' = \frac{\rho_s(1-\gamma_s\Delta T)}{\rho_l(1-\gamma_l\Delta T)} = f(1+(\gamma_l-\gamma_s)\Delta T)$

$\Rightarrow$ $\boxed{f' = f[1+(\gamma_l-\gamma_s)\Delta T]}$

COMPENSATED PENDULUM: An equilateral frame is made from two different types of metals, find $\alpha_1 : \alpha_2$ if the time period of oscillations of pendulum about AB does not change with small changes in Temperature.



Solution: If depth of point C below D does not change then our objective will be met.

Now $DC^2 = BC^2 - DB^2 \Rightarrow h^2 = [2\ell(1+\alpha_2\Delta T)]^2 - (\ell[1+\alpha_1\Delta T])^2$

$\Rightarrow h^2 = 4\ell^2(1+2\alpha_2\Delta T) - \ell^2[1+2\alpha_1\Delta T]$

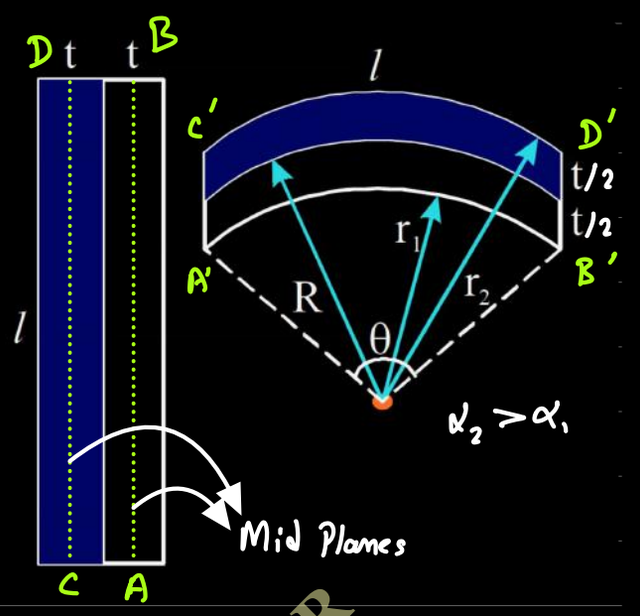
$\Rightarrow h^2 = \underbrace{4\ell^2-\ell^2}_{h^2} + \ell^2\Delta T[8\alpha_2 - 2\alpha_1]$

$\Rightarrow 8\alpha_2\Delta T - 2\alpha_1\Delta T = 0 \Rightarrow \alpha_1 = 4\alpha_2$ Ans.

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BIMETALLIC STRIP: In this arrangement, two metal strips with different expansion coefficients are joined together. Upon heating such a composite strip, it bends due to different lengths of mid planes of two strips.

If R is the radius of interface, then the radii of mid plane of two sticks are:



$r_1 = R - t/2$ and $r_2 = R + t/2$ respectively. Now by thermal expansion:

$$(R - t/2)\theta = l_0(1 + \alpha_1 \Delta T) \quad (R + t/2)\theta = l_0(1 + \alpha_2 \Delta T) \Rightarrow \frac{R - t/2}{R + t/2} = \frac{1 + \alpha_1 \Delta T}{1 + \alpha_2 \Delta T}$$

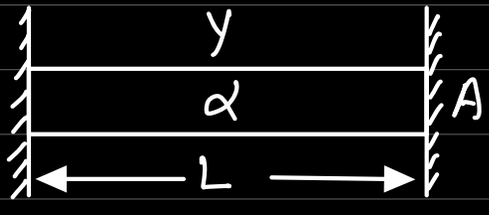
Now (componendo Dividendo $(D+N/D-N)$) $\Rightarrow \frac{2R}{t} = \frac{2 + (\alpha_1 + \alpha_2)\Delta T}{(\alpha_2 - \alpha_1)\Delta T}$

Now $(\alpha_1 + \alpha_2)\Delta T << 2 \Rightarrow \boxed{R = \frac{t}{(\alpha_2 - \alpha_1)\Delta T}}$

For Brass and Iron Strip, $\alpha_B > \alpha_{Fe}$ $\Rightarrow$ Brass is on convex side on heating.

THERMAL STRESS: When a body is not allowed to expand (contract) freely by placing external constraints, the body develops elastic stresses, it behaves as if its natural dimension was the freely expanded dimension, and brought to the given shape through external loading from constraints.

Example: A rod of natural length L just fits between two rigid walls. Find the stress developed in the rod when heated through a temperature ΔT. Also find the force with which rod pushes the walls apart.



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Sol: Natural length: $L_n = L(1+\alpha\Delta T)$

Actual length: $L_a = L$

$\Rightarrow$ Compression: $\Delta L = L_n - L_a = L\alpha\Delta T$

Strain: $\epsilon = \frac{\Delta L}{L_n} \approx \frac{\Delta L}{L} = \alpha\Delta T$

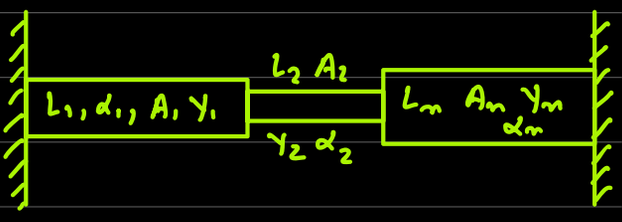
(Note that for sake of convenience we divided by L_a instead of L_n which is OK if expansion is small. It leads to a simpler expression).

Now $\sigma = Y\epsilon = Y\alpha\Delta T \Rightarrow$ $\sigma = Y\alpha\Delta T$ ans.

Now Force $= \sigma \times A = YA\alpha\Delta T$ ← Force pushing Walls.

n Rods In Series Between Rigid Walls:

Consider n rods connected in series between two rigid walls so that they just fit without any stress at some temperature and we wish to find compressive force in each rod, when the system is heated through a temperature ΔT.



→ Recall that Rods behave like springs of constant $k = YA/L$ Since all rods are in series, they must each have same compressive force.

→ Since the rods are restrained between rigid walls, net thermal expansion must be exactly balanced by net elastic compression.

→ Let compressive force in each rod be F, $\Rightarrow$ compression in i^{th} rod $= x_i = F/k_i = FL_i/Y_iA_i$ $\Rightarrow$ total cumulative compression:

$$X = \Sigma x_i = F\Sigma(L_i/Y_iA_i)$$

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Now total thermal elongation Assuming not restraint:

$$\Delta l = \Sigma L_i \alpha_i \Delta T_i, \quad \text{Now by our argument:}$$

$$X = \Delta l \Rightarrow F \Sigma L_i/Y_i A_i = \Sigma L_i \alpha_i \Delta T_i$$

$$\Rightarrow \boxed{F = \frac{\Sigma L_i \alpha_i \Delta T_i}{\Sigma (L_i/Y_i A_i)}}$$

CHANGE IN LENGTH OF ith ROD: Net change in length can now be found by superposing thermal expansion and elastic compression in each rod.

$$\boxed{\Delta l_i = \left(L_i \alpha \Delta T_i - \frac{F L_i}{Y_i A_i}\right)}$$

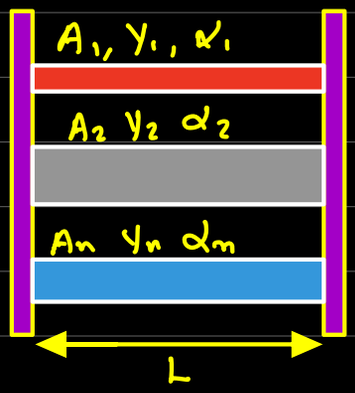
SHIFT IN AN INTERFACE: Total displacement of any interface can be found by adding the changes in lengths of all rods to the left of it (or right of it).

→ A parallel analysis can be done for the case of thermal contraction upon decreasing the temperatures and similar expressions can be derived for case of cooling the system. (Consider ΔT_i as a signed value, +ve for increasing temperature and -ve for decreasing temperature).

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n RODS WELDED BETWEEN MOVABLE RIGID PLATES:
(Bending Prevented by External Couples)

Consider n rods each of length L welded between two rigid but movable plates (Bending prevented by external couples). We wish to find the total elongation in L and individual stresses in rods.



→ The stresses (and forces) in rods could be compressive or tensile (infact for force balance on any plate, some rods must be in tension and some in compression). For ease of analysis we shall consider stress to be a signed value (+ve if tensile and negative if compressive). Then we can simply take the Algebraic sum of all the elastic forces on plates to be zero.

→ Natural length of ith rod : $L_N = L(1+\alpha_i \Delta T_i)$

Now let actual length of All rods be L_A

$\Rightarrow$ Stretch in i^{th} rod $\Delta L_i = L_A - L_N = L_A - L(1+\alpha_i \Delta T_i)$

$\Rightarrow$ Strain in i^{th} rod : $\epsilon_i = \dfrac{L_A - L(1+\alpha_i \Delta T_i)}{L}$

$\Rightarrow$ stress in i^{th} rod : $\sigma_i = \dfrac{Y_i [L_A - L(1+\alpha_i \Delta T_i)]}{L}$

$\Rightarrow$ Force in i^{th} rod : $F_i = Y_i \left[\dfrac{L_A - L(1+\alpha_i \Delta T_i)}{L}\right] \times A_i$ ①

Now For force balance, $\Sigma F_i = 0$

$$\Rightarrow \frac{1}{L}\left(\Sigma Y_i A_i [L_A - L(1+\alpha_i \Delta T_i)\right) = 0$$

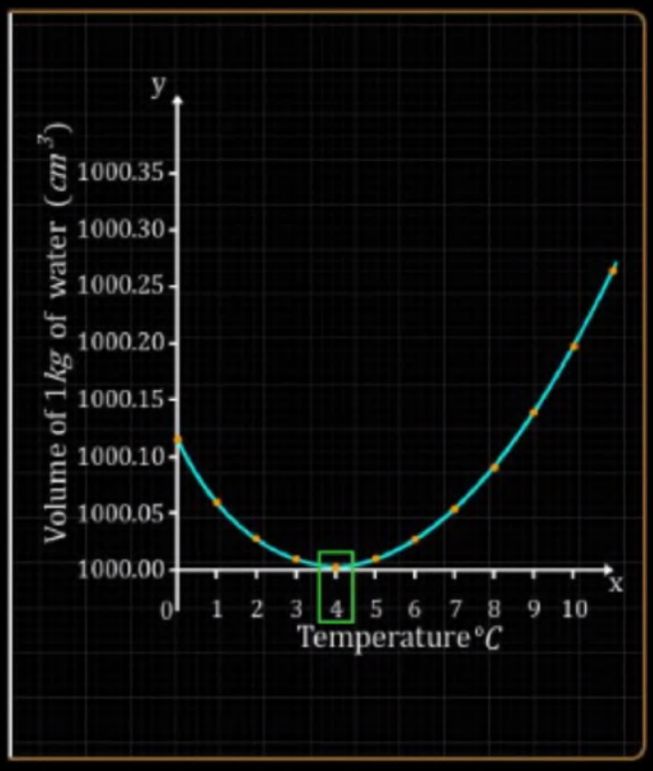
$$\Rightarrow L_A \Sigma Y_i A_i = L \Sigma Y_i A_i (1+\alpha_i \Delta T_i)$$

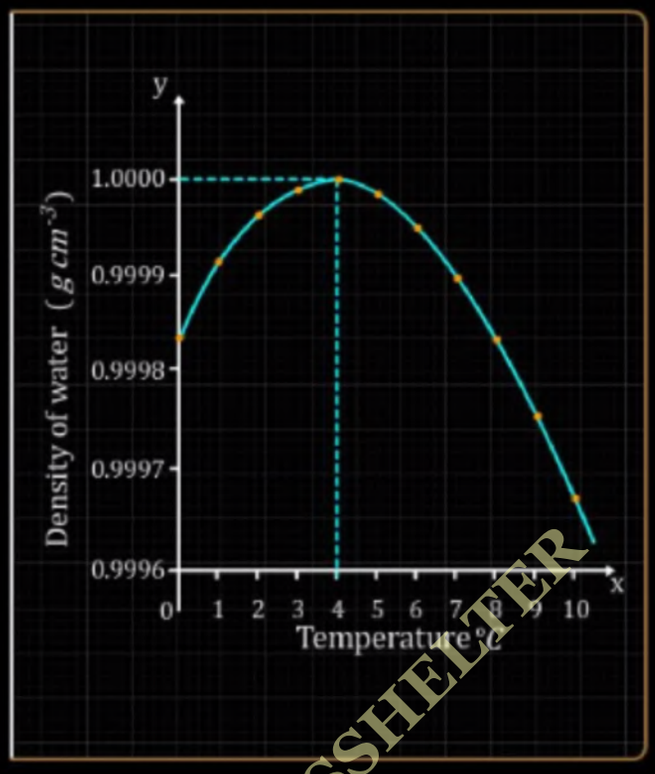
$$\Rightarrow \boxed{L_A = L \frac{\Sigma Y_i A_i (1+\alpha_i \Delta T_i)}{\Sigma Y_i A_i}} \quad ②$$

Now individual forces can be calculated by putting this L_A in equation ①.

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Anomalous Expansion Of Water:





Cause of Anomalous Expansion: In the ice phase, the water molecules form a crystalline structure, however upon melting, the crystal structure begins to break, and amorphous nature begins to take over, thus increasing the density (Somewhat like crumpling a box made of corrugated cardboard). The density becomes maximum at 4°C and as the temperature increases further, the intermolecular spacing increases due to thermal vibrations taking over and thermal expansion starts.

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Example: Stress In A Spherical Tank:

A rigid spherical tank of radius R and small wall thickness t contains a liquid of coefficient of volumetric expansion coefficient γ. Calculate increase in stress in its wall if the temperature increases by ΔT. Bulk Modulus of liquid is B.

Solution: Natural Volume = $V_0(1+\gamma\Delta T)$, Actual Volume = V_0

$\Rightarrow$ Compression: $\Delta V = V_0\gamma\Delta T$

Now $B = \dfrac{\Delta P}{|\Delta V/V|} \Rightarrow \Delta P = B\left|\dfrac{\Delta V}{V}\right| = B\gamma\Delta T$

Also excess pressure is related to extra stress as: $\Delta P = (2\sigma t/R)$

$\Rightarrow \sigma = \dfrac{R}{2t} \times \Delta P = (B\gamma R\Delta T)/2t$ Ans.

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Q: Repeat the previous problem if Bulk modulus of liquid were infinity but vessel had a Young's Modulus Y and negligible coefficient of thermal expansion.

Solution: Infinite Bulk Modulus implies that the liquid is incompressible
$\therefore$ vessel must expand to new volume of liquid.

Now: $\Delta V = V_0 \gamma \Delta T$, Also $V = \frac{4}{3}\pi R^3 \Rightarrow \frac{\Delta V}{V} \approx \frac{3\Delta R}{R} = 3\epsilon$

$$\Rightarrow \epsilon = \frac{\gamma \Delta T}{3} \Rightarrow \sigma = Y\epsilon = \frac{Y\gamma\Delta T}{3} \text{ Ans.}$$

EXAMPLE:

Left end of a rod is fixed and initial length of the rod is l_0 at $0^\circ C$. Now a linear temperature variation is established with left end still at $0^\circ C$ & right end at $\theta^\circ C$. Find:



(a) Shift in CM of the rod

(b) Shift in a point originally located at a distance x from left end. Coefficient of linear expansion is α (Constant).

Solution: Consider an element at distance x (unstretched) and thickness dx. Let dy be its elongation.

$$\Rightarrow dy = \alpha\, dx \times \left(\frac{\theta x}{l_0}\right)$$

Shift in element located x is summation of elongation in all elements up to x. Now integrating:

$$y = \frac{\alpha\theta}{l_0}\int_0^x x\,dx = \frac{\alpha\theta x^2}{2l_0} \text{ Ans.}$$

$$\text{Now shift in CM} = \frac{\sum m_i y_i}{\sum m_i} = \frac{1}{m}\int_0^{l_0} \frac{m}{l_0}dx \times \frac{\alpha\theta x^2}{2l_0} = \frac{\alpha\theta l_0}{6} \text{ Ans.}$$

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